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AgriForecast Kenya

Kenya Agricultural Market Insights and Forecasting System with Sentiment Analysis

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Turn Over

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Introduction

Agriculture contributes 33% of Kenya’s GDP and employs 70% of the rural workforce, yet price volatility, market inefficiencies, and climate change pose major challenges. Our goal is to develop a data-driven forecasting system for commodity prices leveraging historical price data, machine learning, and sentiment analysis.

Why Price Forecasting?

- Helps farmers & traders maximize profits
- Supports policymakers in market stabilization
- Aids investors in decision-making

Objectives

- Analyze commodity price trends
- Predict future price movements
- Assess the impact of public sentiment

Would you trust a price forecast from a machine learning model, or are your grandmother’s “weather knee” predictions still superior?

Business Understanding

Problem Statement

Kenya’s agricultural sector faces price fluctuations of up to 32%, impacting farmers, traders, policymakers, and consumers. Accurate price forecasts enable better planning, investment decisions, and market stability, ultimately enhancing efficiency, profitability, and food security.

If you had real-time price forecasts, what’s the first thing you’d do—buy, sell, or start a TikTok channel about it?

Objectives

Develop a forecasting system that provides actionable insights into agricultural commodity prices by:

- **Assessing Price Fluctuations** – Analyze historical data to identify trends and volatility.
- **Predicting Market Trends** – Use machine learning to forecast future price movements.
- **Incorporating Sentiment Analysis** – Track public sentiment from social media to understand market impact.

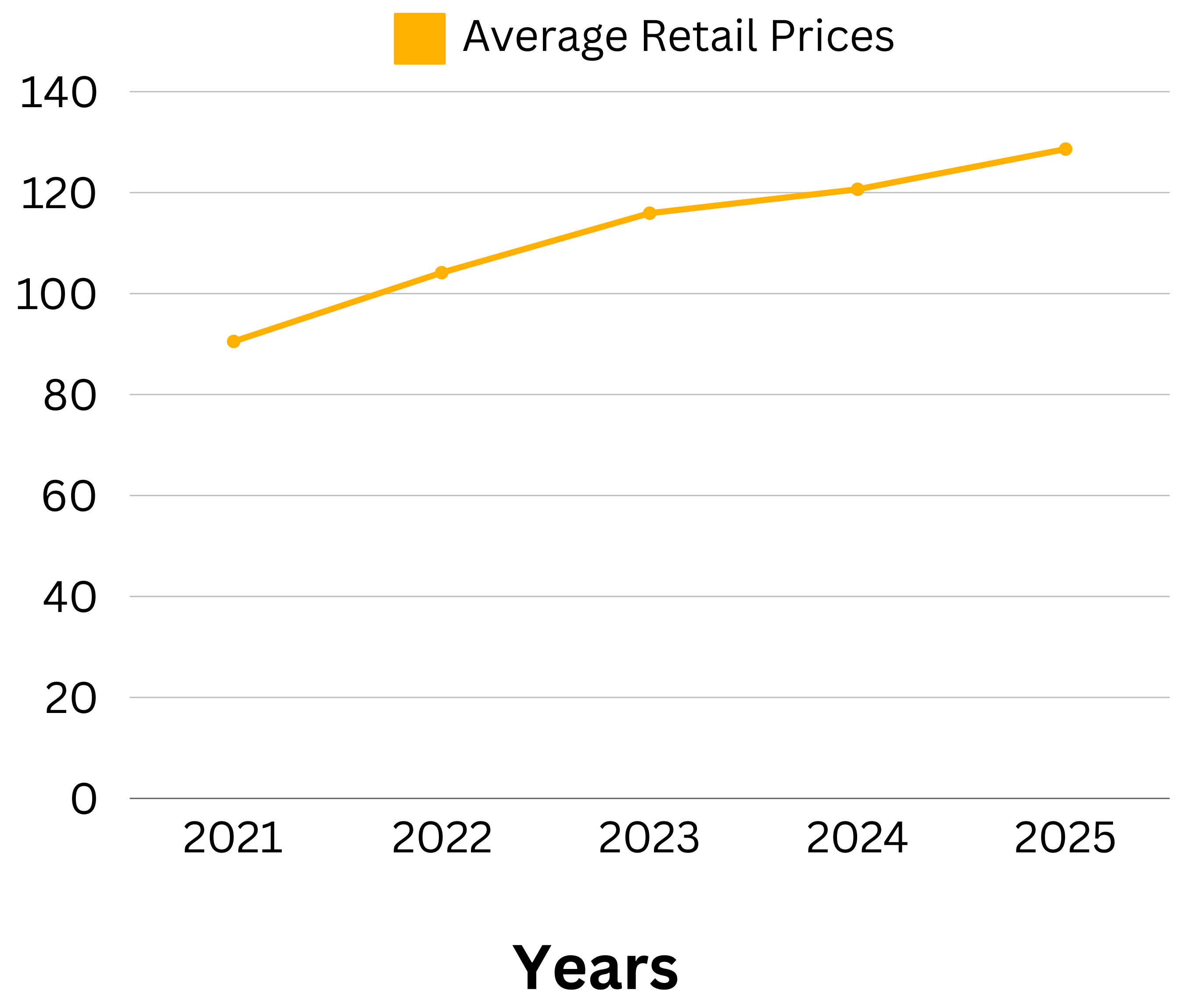
This project empowers farmers, traders, and policymakers to make informed decisions and enhance market stability.

Data Sources & Understanding

This project uses data from reliable sources, including KAMIS for commodity prices, geographical data for market locations, and Twitter sentiment for public perception. The dataset contains 288,478 records detailing commodity types, prices, and supply volumes. Inconsistencies include 5% missing price data, 25% incomplete location entries, and sentiment data requiring preprocessing for accuracy.

Exploratory Data Analysis

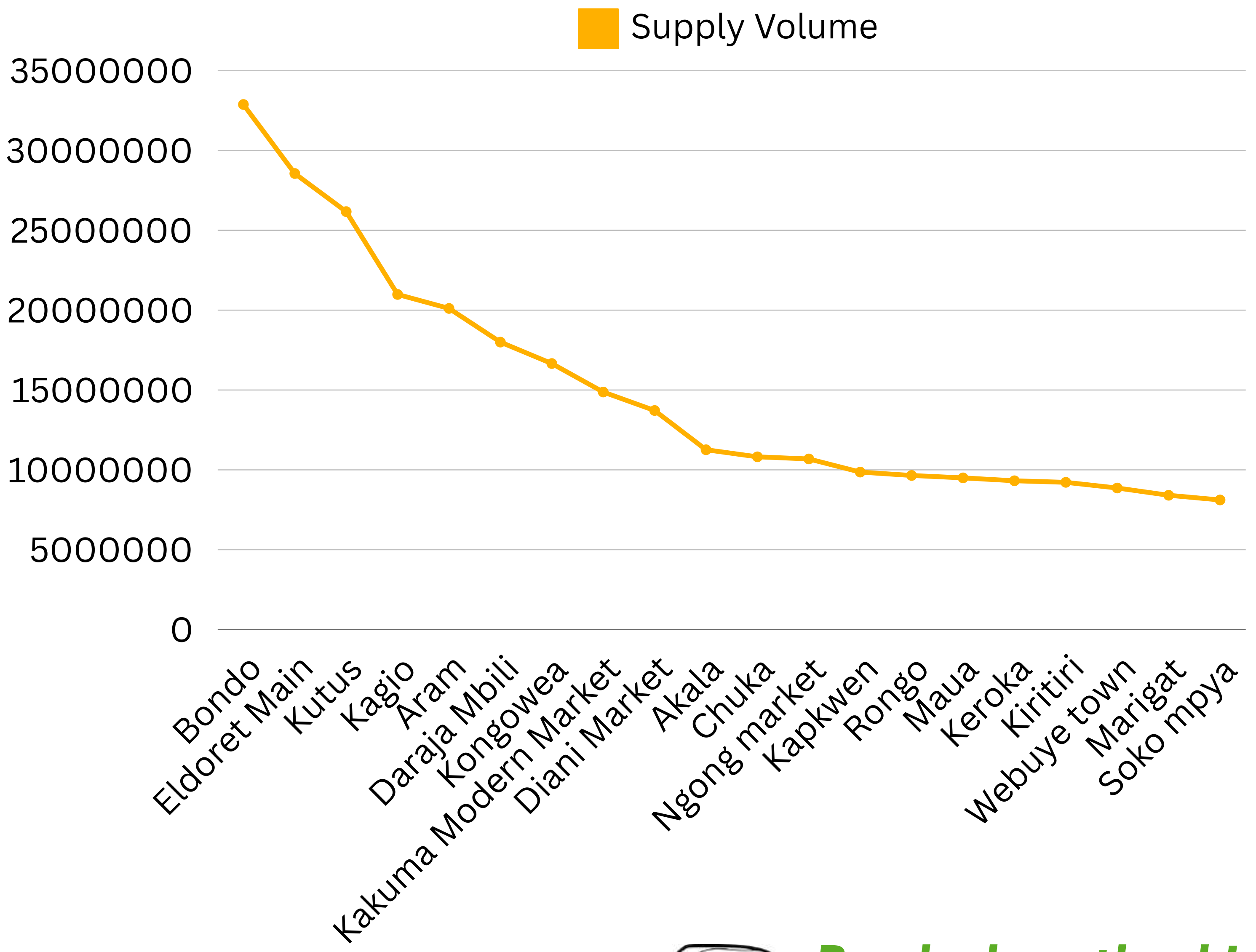
Commodity Price Trends Over Time

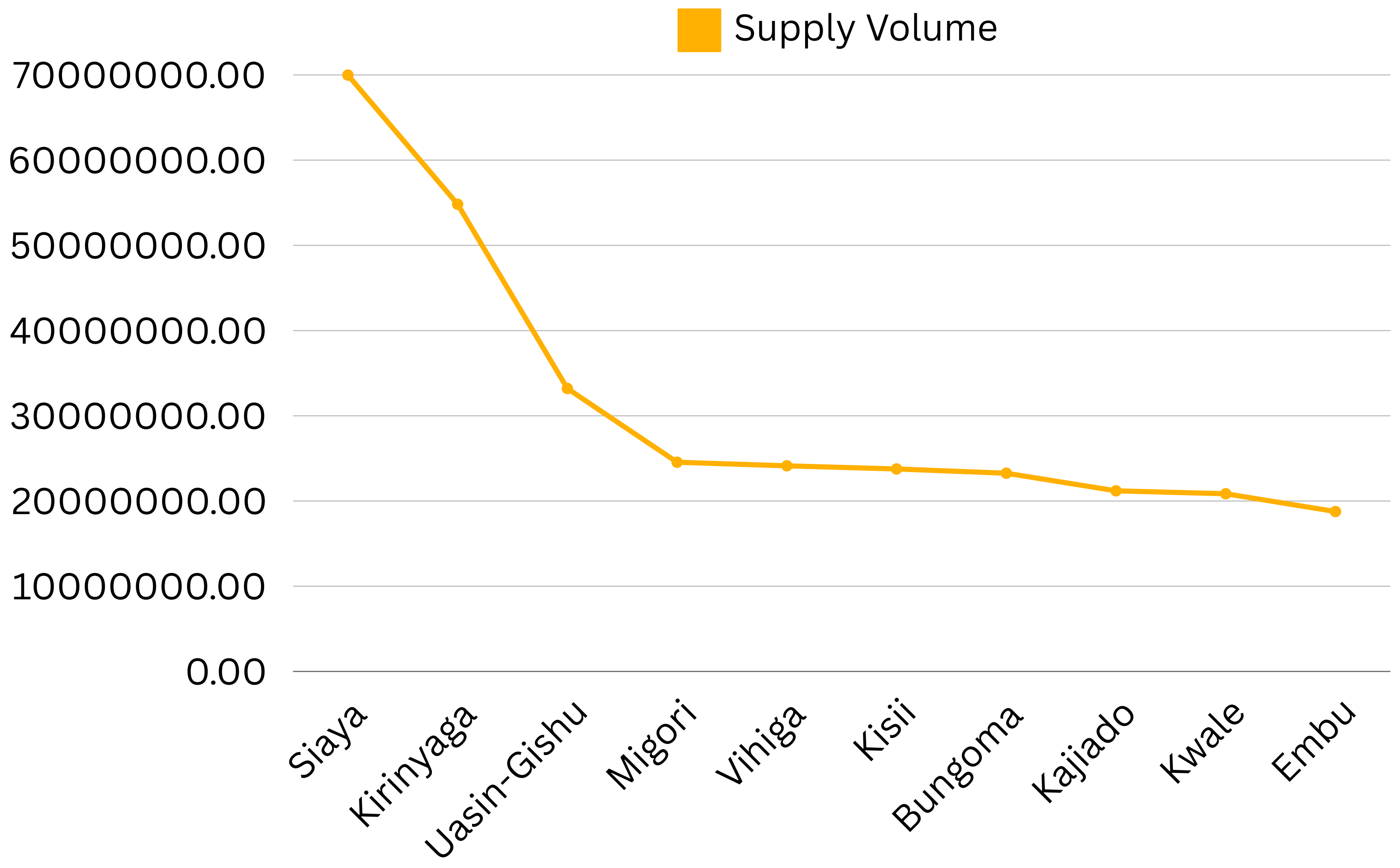


Explanation

Kenyan food prices follow seasonal patterns, peaking between January and April as farmers retain produce for seeds. Prices drop from April to June with the onset of harvests and further decline between June and December due to increased agricultural output.
From 2021 to 2025, prices rose steadily, with a sharp spike in 2022 linked to the Russia-Ukraine conflict. Ongoing global disruptions, climate change, and economic factors continue to influence market volatility.

Regional Variations in Prices



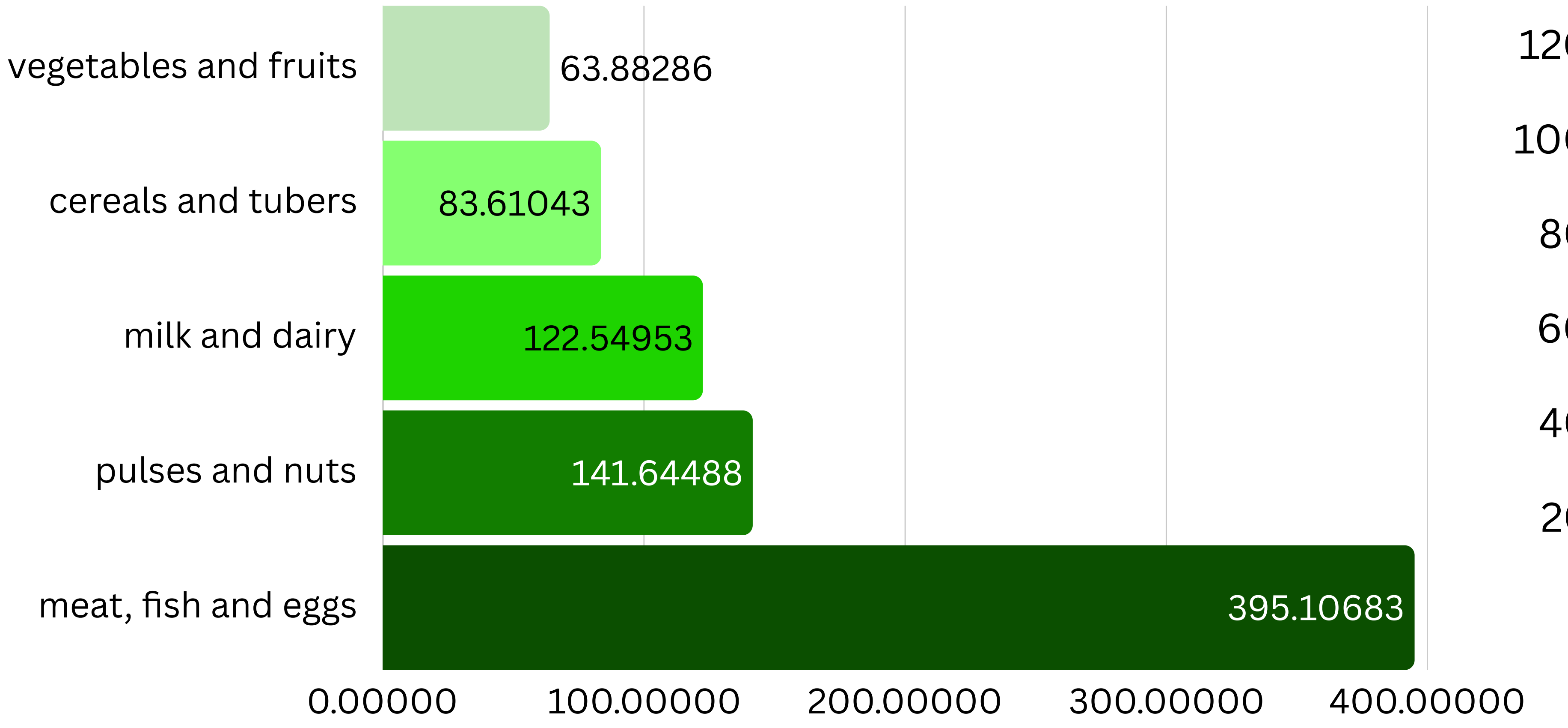


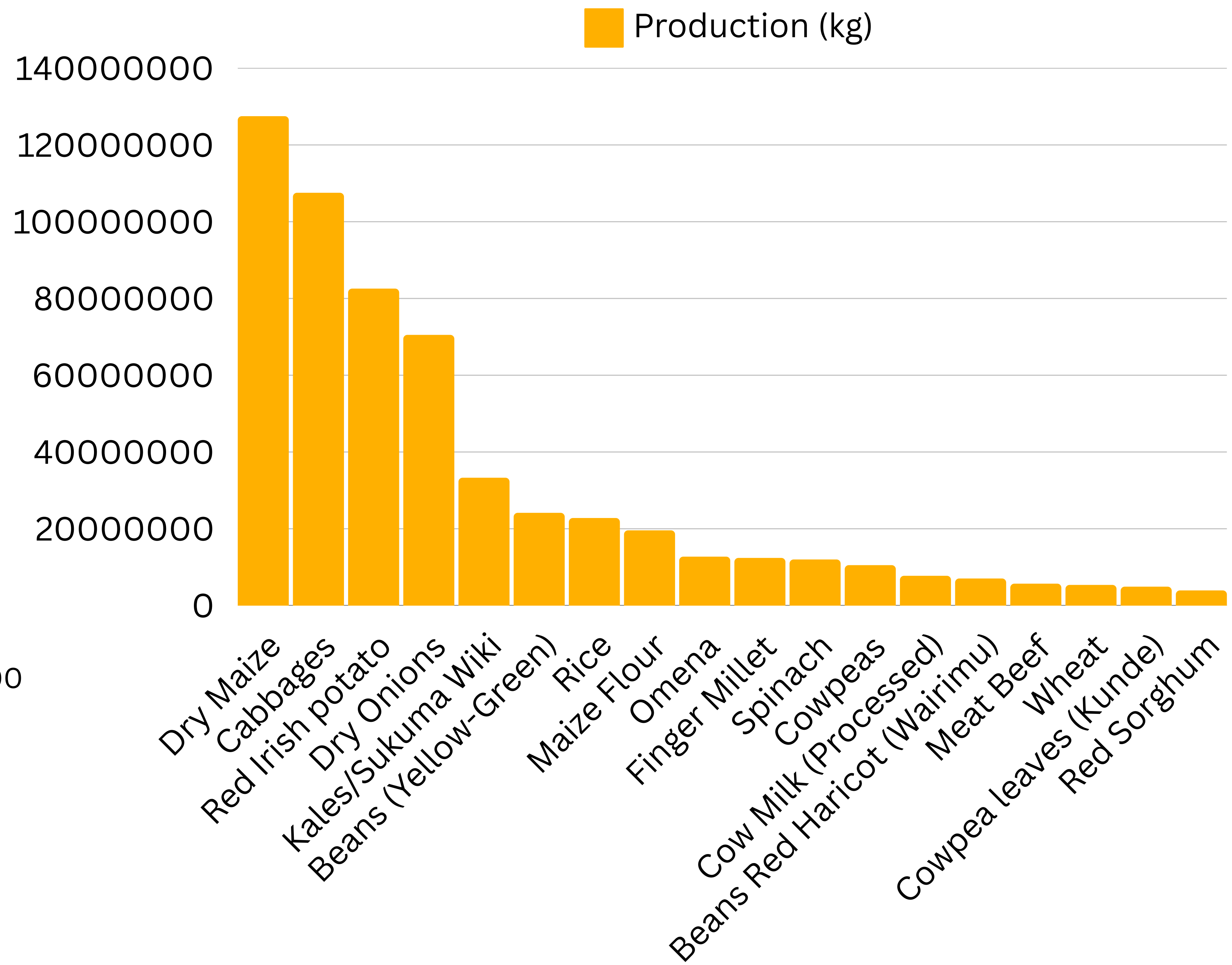
Bondo has the highest supply volume—do they know something the rest of Kenya doesn't?

Explanation

Bondo leads in supply volume, followed by Siaya, Kirinyaga, and Uasin Gishu, highlighting key trade hubs. Urban centers like Mombasa and Kisumu drive demand, while arid regions like Turkana and West Pokot still contribute significantly. This distribution reflects regional production strengths and market dynamics.

Product Compositions in Markets





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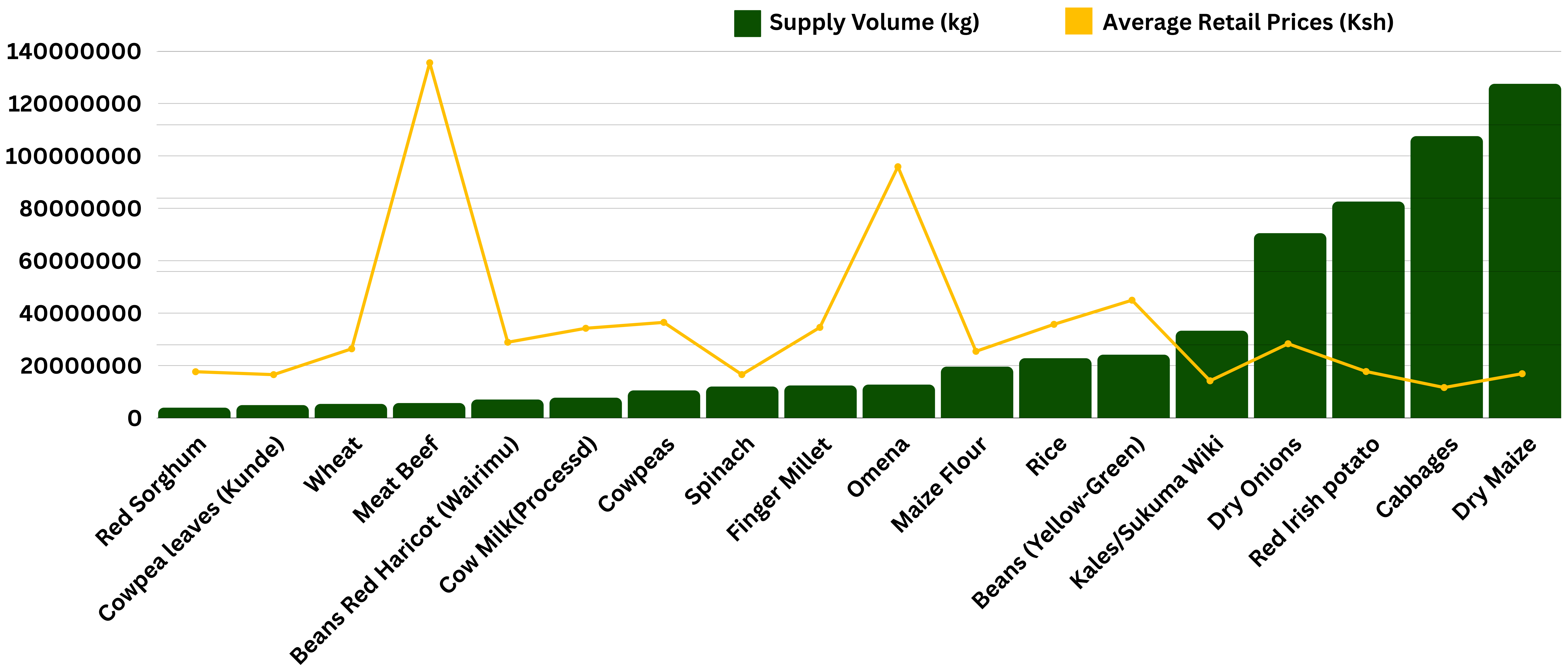
Explanation

Meat, fish, and eggs have the highest prices, while staples like maize and beans remain affordable, driven by supply chain costs and seasonal availability. Dry Maize dominates supply, followed by cabbages and potatoes, while wheat and beef have lower volumes, impacting market prices and food security.

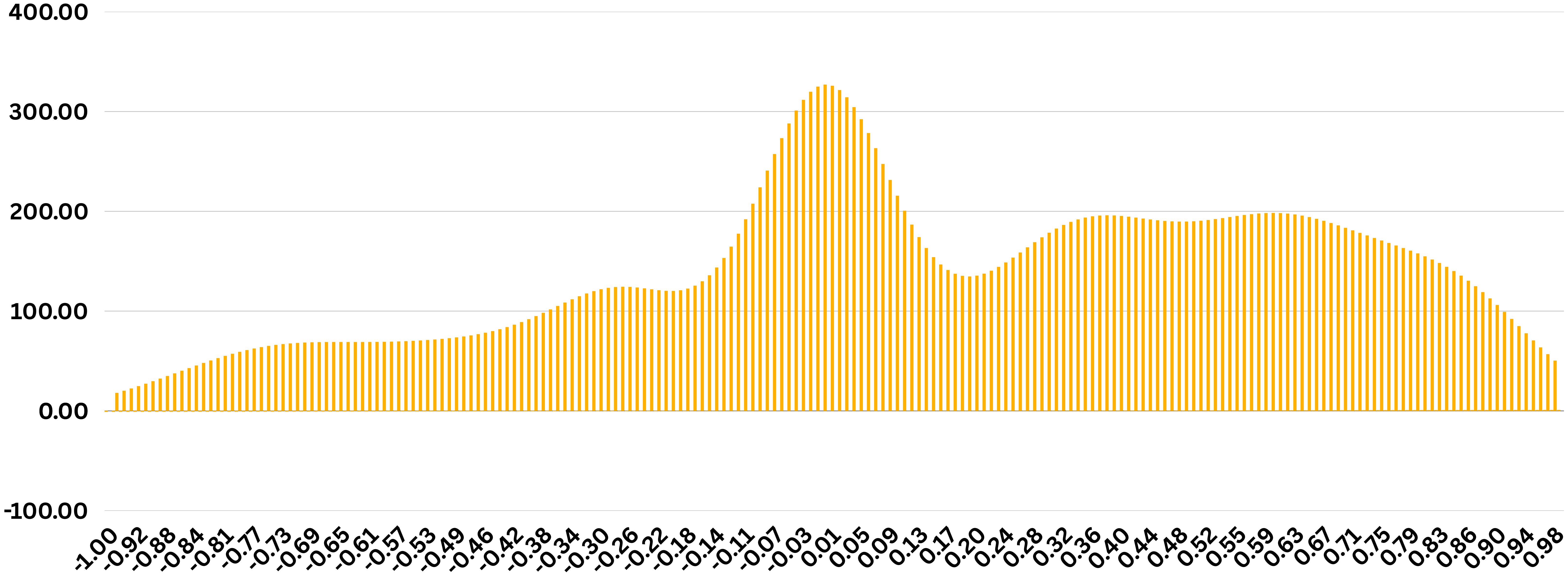
This is best illustrated in the following graph:

Relationship between Supply Volume and Product Prices



Sentiment Analysis

If Twitter sentiment can predict market trends, does that mean complaining online is now a valuable economic indicator?



Explanation

Overall Sentiment

- Average sentiment score: 0.17 (mildly positive).
- Neutral sentiments dominate, with fewer negative opinions.

Sentiment Distribution

- Most posts have positive sentiment (peak around +0.5).
- Negative sentiment is less common but spikes during price hikes.

Sentiment Over Time

- Fluctuations indicate market volatility.
- Timely monitoring can improve decision-making.

Engagement & Sentiment

- Minimal correlation between sentiment & retweets/likes.
- Highlights the complexity of social media interactions.

Machine Learning

We aimed to forecast commodity prices using machine learning and analyze public sentiment to understand its impact on market trends.
We forecasted commodity prices using XGBoost (94.5% accuracy) and LSTM (87.4%) while VADER (Calculating tweet sentiments) analyzed public sentiment. These models provided data-driven insights for better market decisions

Our model is 87% accurate—would you trust it with your money, or are you still waiting for a "100% no-fail" guarantee?

Key Findings

Seasonal Price Trends

- Prices peak Jan–April (planting) & drop June–Dec (harvest).
- 2022 spike linked to global shocks.

Regional Price Differences

- Urban areas (Nairobi, Mombasa) → Higher prices.
- Remote areas (Turkana, West Pokot) → Transport costs.

Supply & Price Correlation

- High supply (Dry Maize) → Stable prices.
- Low supply (Meat, Wheat) → Price volatility.

Machine Learning Performance

- XGBoost (94.5%) & LSTM (87.4%) predict prices well.
- Helps farmers, traders, policymakers.

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CONCLUSION

- Prices follow seasonal patterns. They peak from January to April and drop from June to December due to harvests. The 2022 spike was caused by global disruptions.
- Regions show price differences. Urban and arid areas have higher prices due to demand. Remote areas face high transport costs, raising food prices.
- Supply volumes affect their prices.
- Machine learning models work well. XGBoost (94.5%) and LSTM (87.4%) predict prices accurately. VADER classifies sentiment effectively

If sentiment analysis shows mostly neutral opinions, does that mean Kenyans are just tired of complaining?

Cheers!

Thank You